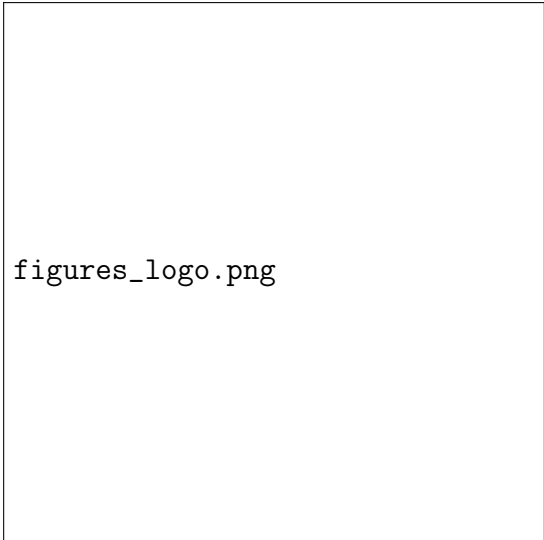




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CRITICAL RESEARCH REVIEW ARTICLE

THE ROLE OF DIGITAL FORENSICS IN CYBERCRIME INVESTIGATIONS

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Abstract

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The abstract should include:

- Background/context of the selected research topic
- Purpose of the review

- General approach used to select and analyse the research articles
- Major findings from the reviewed literature
- Key research gaps identified
- Overall conclusion

Recommended length: 150–250 words.

Keywords: Digital Forensics; Cybersecurity; Investigation; Multimedia forensics; Cybercrime; Biometric; Data privacy

Contents

1 Introduction

1.1 Background

As digital technology has experienced rapid evolution and expansion throughout the years, it has slowly integrated itself into most aspects of our own lives, from communications to our governance. Naturally, this boon that provides us convenience and connectivity will bring upon malicious actors that will misuse them to commit cybercrimes, ranging from small scaled crimes like identity theft and cybertheft of financial and card payment data to much larger scale ones such as cyberespionage and ransomware attacks.

As such, traditional methods of investigation have been proven to be ineffective in combating and solving these crimes. Hence, digital forensics as a discipline has seen a rise within the fields of cybersecurity and law enforcement to ensure that the malicious actors will have their deserved justice dealt to them alongside learning their methods to better suppress the surge of cybercrime.

1.2 Problem Statement

Almost every part of daily life now runs through a computer or a phone, and crime has followed right along with it. Money gets stolen, systems get broken into, and all of it can happen from the other side of the world. Digital forensics — the work of digging through devices and networks for proof of what happened — has become central to how law enforcement operates. That said, the field is under real strain right now.

For a long time, the basic strategy was straightforward: keep a list of known bad signatures and block anything that matched. That worked fine when attackers were unsophisticated. They aren't anymore. Advanced Persistent Threats, for example, are designed to slip past defenses and remain undiscovered for lengthy periods of time, silently removing important data from a network without raising any alarms. The scope of the problem is difficult to overestimate – cybercrime is expected to cost the global economy 10.5 trillion dollars by 2025, yet discovery is sometimes excruciatingly slow: one research estimated that the average time to find a breach was 212 days.

Part of the difficulty stems from the fact that digital forensics is a collection of disparate fields. Network forensics works with traffic, mobile forensics with phones and tablets, and then there's cloud, database, IoT, and multimedia forensics, each with its own set of tools and standards. A case that involves more than one of these areas requires investigators to piece together methodologies that were never intended to function together, slowing things slowly and introducing opportunity for mistake.

Criminals aren't making this easier, either. Encryption scrambles communications. Steganography hides malicious code inside what looks like an ordinary image. Fileless virus operates totally in memory and leaves no trace when the system reboots. Some players are even using blockchain to hide their actions. Even when investigators uncover anything, it does not always make it to court – shoddy paperwork, uneven processes, and contradicting legislation across jurisdictions are major reasons solid evidence is tossed out.

Artificial intelligence is frequently presented as the solution. It's easy to see why: machine learning is genuinely good at finding patterns buried in massive datasets and flagging behavior that a human analyst might miss, and there's evidence that AI-powered tools outperform older methods in terms of detection rate and false positives. But AI

brings its own baggage. These models are frequently black boxes, so it’s not always clear why a system flagged what it flagged. They can be manipulated with carefully designed adversarial inputs. They require massive training datasets, which are not always accessible, as well as significant computational capacity to function.

Then there’s the smart-device problem. Homes are full of smart speakers, locks, thermostats, TVs; factories run thousands of interconnected sensors. All of this results in a massive, jumbled volume of data in inconsistent forms, dispersed among devices, cloud servers, and local storage. Figuring out where the relevant evidence even lives, let alone extracting it and proving it hasn’t been tampered with, is genuinely difficult in this kind of distributed environment.

The legal landscape doesn’t help. Privacy laws vary greatly by country; some countries openly share evidence while others do not, and permissible investigation tactics range from one another. Investigators must operate within this patchwork while adhering to guidelines such as GDPR and CCPA, and a misstep here might torpedo a case completely.

Despite years of research in this area, there is still no cohesive method that tackles all of these issues simultaneously. Standardized testing techniques for forensic tools are sparse, multi-domain datasets are scarce, and technologies produced by various manufacturers frequently do not work together. Furthermore, new dangers such as deepfakes and adversarial assaults continue to emerge and necessitate their own solutions.

1.3 Objectives of the Review

This study aims to assess where digital forensics is today: what has been solved, what has not, and where the profession appears to be headed — with a focus on how AI and similar technologies might boost investigations and protect evidence integrity.

More specifically:

First, we look at current practice in six domains: network, mobile, database, IoT, cloud, and multimedia forensics, highlighting the key issues, the tools and methodologies used, and the largest gaps in each.

Second, we evaluate how well AI-based methods function in practice, focusing on deep learning, federated learning, and anomaly detection to identify where these approaches succeed and where they fail .

Third, we examine the problems provided by new threats such as APTs, encryption, anti-forensic tactics, steganography, and AI-generated media, as well as potential responses .

Fourth, we look at the legal and ethical frameworks that shape the field, such as how courts assess the admissibility of digital evidence, chain-of-custody requirements, privacy protections, and the challenges that come with cross-border investigations.

Fifth, we present a cross-domain framework that combines federated learning with context-aware policy enforcement, which is intended to function across hybrid cloud, IoT, and corporate contexts while addressing the constraints of current single-domain techniques.

Finally, we highlight areas for future research, including explainable AI, quantum-resistant encryption, blockchain-based evidence tracking, privacy-preserving threat intelligence sharing, and automated attack attribution.

1.4 Research Questions

RQ1: What are the predominant methodologies and tools used in cybercrime investigations?

RQ2: What are the implications of artificial intelligence use in digital forensics?

RQ3: What are the methodological limitations identified in the current studies on digital forensics?

RQ4: What areas of digital forensics warrant further research?

2 Review Methodology

The methodological process of conducting the literature review was by following an ordered, systematic approach that ultimately results in synthesis of new information based on our chosen research articles.

To begin, each article was thoroughly read to form an accurate context and understanding of the study conducted. Before extraction of any information, the aim was to fully comprehend the text, with any unfamiliar terms or procedures investigated to allow for an accurate depiction of the content.

Subsequently, a second iteration of reading the text follows, with the main purpose to extract key findings, methodologies, and general points. The information was compiled into a comparative table in which summarized each paper for easy reference.

All methodologies per paper were scrutinized by analyzing how the study was conducted, the tools it used and the type of findings it obtained. The disadvantages and advantages were recorded and identified for further analysis.

Next, like articles in terms of area of study, general conclusions and method of research were grouped together, with an identification of a general theme and their limitations. For each theme, a critical analysis was executed.

After recognizing the groups, a draft was created to organize synthesized points to ensure the review follows a coherent order and is framed with an intuitive structure.

2.1 Literature Search Strategy

To ensure the analysis we conducted are within the scope of our research and provides comprehensive depth, it is imperative to employ necessary techniques and paper selection for proper evaluation.

Firstly, a simple protocol was followed when selecting papers such that it is sourced explicitly using from trusted sources such as IEEE Xplore, ScienceDirect, IAES International Journal, arXiv, NUML International Journal. These academic databases were chosen for their viability to be cited, their coverage and for their contributions to digital forensics in cybersecurity.

A combination of keywords were used as search terms such as “digital forensics,” “cybersecurity,” “investigation,” “artificial intelligence in digital forensics.” These words along with “AND” and “OR” boolean operators are used to refine the search and identify available and relevant papers.

Once the search had narrowed, papers were selected based on an initial screening based on title relevance to our subject, and were viewed in full-text to skim through its content.

Following the compilation of papers, an inclusion and exclusion criteria were used to determine whether it would be included or discarded.

2.2 Inclusion and Exclusion Criteria

To ensure the selection of research articles are of high quality and relevant to our research, a systematic inclusion and exclusion criteria were followed strictly.

Inclusion criteria:

1. Relevance: The paper must continuously be relevant to the current subject and contains at least one aspect of digital forensics.
2. Publication Date: Published between 2021 and 2026 to ensure that the studies discussed are updated and reflect current issues.
3. Page Length: Its pages were a minimum of 6 pages in length to ensure wide depth coverage on the subject.
4. Accessibility: The full-text is available at all times to be viewed for proper analysis.
5. Complete Information: Bibliographic information are complete and readily available for proper citation and referencing.
6. Coverage: The coverage of its subject is diversified and allows for new research areas to develop.
7. Research Contribution: the studies must have its own original findings through experiments or have a concise discussion related to digital forensics.

Exclusion criteria:

1. Published before the year 2021.
2. Page length is less than 6 pages.
3. Full-text is not readily available to view.
4. Insufficient bibliographic information for proper referencing.
5. Its research methodology is not readily disclosed and transparent.
6. Its title relevance to our subject were determined out-of-scope.
7. In a language other than English.

2.3 Article Selection

Following the initial compilation of research articles through title relevance, papers were rejected based on its page length being less than 6 pages, its publishing before 2021, and its insufficient research, amongst others.

Once compiled, each paper was meticulously examined through abstract inspection and the content of the main body's key findings and points. Concluding the article selection, 12 papers were selected based on the viability of adhering to the inclusion and exclusion criteria and was taken for literature review and further analysis.

3 Literature Review

3.1 Theme 1: Standardised Practices and Advancements of Digital Forensics

Through the rampant evolution of cybercrime in its sophistication of execution alongside the current digital age allowing access for various methods to execute cybercrime, digital forensics as a field within computer science has gone through many advancements to ensure that cybercriminals will face their due justice.

To better evaluate Digital Forensic’s role in investigating and solving cybercrime, it would be apt to first understand the standardised methods of which the practitioners of digital forensics utilise in order to obtain the proper digital evidence for the committed cybercrime, and then delve into the various advancements made towards improving digital forensics’ own practices as a whole.

Table 2 showcases research articles which align with the current theme of Standardised Practices of Digital Forensics, alongside their methodologies of research.

Table 1: Research Articles for Standardised Practices of Digital Forensics and their Methodology

Research Articles	Primary Objectives	Methodology
Ilyas et al., 2024	To analyze the extent and legal strength of digital forensic evidence within the Indonesian legal system.	Normative: Viewing law as a system of norms. Utilises philosophical (Nature of digital forensics), conceptual (Test and analyse concepts), and case (study and analyze electronic crime cases) approaches.
Hassan Shah Khan, 2025	To evaluate the effectiveness of current digital forensic strategies and the possibility of enhancing them	Qualitative: Comparative analysis of digital forensic tools alongside examination of high-profile cybercrime case studies.
Sharma, 2024	To investigate the adherence to legal standards and reporting procedures in forensic investigations.	Quantitative Content Analysis: Literature review of 40 reports from a quasi-experiment
Arif et al., 2025	To evaluate the challenges of digital forensics the existing methodologies face	Systematic Literature Review (SLR): Analysis of peer-reviewed publications from 2019 to 2025 across prominent databases.

3.1.1 Comparison of Research Objectives and Scope

The chosen research articles intends to explore the transformation of digital forensics from traditional reactive models to much more reactive and intelligent frameworks.

- **General Evolution and AI Integration** : Research by Vaddi et al., 2024 focuses on the evolution of digital forensics over the past five years, specifically aiming to provide a high-level overview of AI’s potential to solve modern difficulties. Their objective is to guide developers in constructing intelligent, automated tools for effective crime detection across blockchain, cloud, and deep learning (DL) domains Vaddi et al., 2024.

- **Smart Environments and Forensic Readiness:** Alenezi, 2023 shifts the focus toward "smart environments" (homes, cities, and IoT). The primary objective is to survey recent advancements and challenges in retrieving and preserving evidence across a multitude of connected devices, while proposing potential solutions to overcome the lack of standardization in these environments (Alenezi, 2023).

Table 3 showcases research articles which align with the theme of the Advancements of Digital Forensics, alongside their methodologies of research.

Table 2: Research Articles for Advancements of Digital Forensics and their Methodology

Feature	Vaddi et al., 2024	Alenezi, 2023
Research Type	Passive data gathering, literature and statistical analysis	Comprehensive survey
Temporal Focus	Patterns no older than 2018	Recent advancements and possible challenges
Core Methodology	Evaluating long-standing and latest techniques alongside summarising sections to keywords	Examining strengths and limitations of state-of-the-art tools and identifying major challenges and difficulties.

3.1.2 Standardised Investigative Procedures and Workflows

A central theme across the literature is the necessity of a structured, multi-stage forensic process to ensure the integrity and admissibility of evidence. While terminology varies slightly, the core stages remain consistent.

3.1.3 The Forensic Lifecycle Comparison

The Forensic Lifecycle Comparison Standardised practices generally adhere to a flow of identifying, securing, and interpreting data.

- **Ilyas et al., 2024** outlines a four stage model: Identification, Examination, Analysis and Reporting. The article emphasises the digital evidence obtained through the four stage model to be volatile and must be stored carefully for proper presentation of the evidence in a trial to ensure it is not defective in the eyes of the law.
- **Hassan Shah Khan, 2025** instead presents a six-stage model: Identification, Collection, Preservation, Examination, Analysis, and Reporting. Key difference in comparison to the four stage model being the Collection and Preservation stages to ensure the integrity of the collected information, through use of write blockers, regular integrity checks and storage in secure, tamper proof containers
- **Arif et al., 2025** instead suggests a smoother model, consisting of Identification, Preservation, Analysis, Documentation, and Presentation. Their focus is on a high-level theoretical metamodel designed to unify tasks across subdomains like IoT and Cloud forensics to reduce "process redundancies."

3.1.4 Key Procedural Requirements

Specific principles of digital forensics are treated as standardised practice, notably those cited from the Association of Chief Police Officers (ACPO) Ilyas et al., 2024 :

- Non-interference: Digital data is prohibited to be changed before being brought to court.
- Competence: Anyone accessing digital evidence must show a degree of competence alongside understanding of the implications and the relevancy of their actions upon said evidence.
- Documentation: Steps taken on the storage media while its being analysed and investigated must be audited detailedly so that they are able to be replicated exactly with the same results by a third party.

3.1.5 Dataset, Tools, and Technology Comparison

To be able to acquire proper digital evidence, a multitude of tools will be required. One particular research article, particularly Hassan Shah Khan, 2025 manages to compare and contrast multiple various software and hardware solutions, all utilised in different ways both professionally and academically, which is properly displayed in Table 4.

Table 3: Forensic Tools and Use Cases

Category	Tools Identified	Strengths	Limitations
Computer Forensics	EnCase, FTK (Forensic Toolkit)	Comprehensive file systems and metadata analysis	High tool costs, high demand
Mobile Forensics	Cellebrite UFED, MSAB XRY, Oxygen Forensic Suite	Effective data extraction from locked/encrypted mobile devices	Remote-wiping media, device
Network Forensics	Wireshark, NetWitness, Snort, Tcpdump	Real-time traffic monitoring and packet-level analysis	High expertise required, steep learning curve
Cloud/IoT	Magnet AXIOM, IoT Inspector	Specialized for multi-tenant environments and smart device vulnerabilities	Jurisdictional, privacy protocol and ethical

3.1.6 Comparative Analysis of Technical Frameworks and Tools and their Advancements

The articles highlights a transition toward automated, decentralized, and layered forensic architectures.

1. Artificial Intelligence and Machine Learning Both articles agree that AI and ML are essential for managing the volume and complexity of data generated in modern environments.
 - **Architectural Components:** Vaddi et al., 2024 propose a hybrid model consisting of a User Interface (for crime type selection and data entry), a Feature Extraction Module (using OCR and voice recognition), and a Crime Matching Engine. This engine utilizes multi-linear perceptrons, K-nearest neighbor (KNN), and neural networks Vaddi et al., 2024.
 - **Functional Applications:** Alenezi, 2023 categorizes AI/ML applications into image/audio analysis (facial/voice recognition), text analysis (sentiment and keyword identification in chat logs), and behavioral analysis (identifying suspicious patterns in user activity).
2. Cloud Forensics

- **Layered Frameworks:** Vaddi et al., 2024 describes a cloud framework separated into four distinct layers: abstract, information gathering, data storage, and information fusion. This framework utilizes a "voting-k-means" approach for evidence separation Vaddi et al., 2024.
- **Investigation Focus:** Alenezi, 2023 emphasizes the analysis of virtual machine images and the reconstruction of user activity within cloud-based services, highlighting the need for cloud forensic readiness to address data ownership and privacy (Alenezi, 2023).

3. Blockchain and IoT Integrity

- **Chain of Custody:** Vaddi et al., 2024 present a forensic-chain framework for IoT. This utilizes a distributed ledger to record analysis events (Technical Evidence or TEs), ensuring immutability through Merkle trees and simplified Proof of Work (PoW) algorithms Vaddi et al., 2024.
- **Evidence Management:** The blockchain framework uses smart contracts to automatically log every evidence transfer, including timestamps, permission levels, and recipient addresses Vaddi et al., 2024.

4. Network and Memory Forensics

Alenezi, 2023 provides a detailed breakdown of network tools not extensively covered in the AI-centric Vaddi et al., 2024 study:

- **Packet Capture:** Use of Wireshark, Snort, and tcpdump to identify suspicious traffic flows (Alenezi, 2023).
- **Flow Analysis:** NetFlow and sFlow for detecting DoS and DDoS attacks.
- **Memory Forensics:** Analysis of computer memory to uncover malware and network intrusions in IoT and mobile devices (Alenezi, 2023).

3.1.7 Experimental Setup and Data Sources

The research articles utilizes diverse data points to validate their own discovery and findings:

- Ilyas et al., 2024: Used data from the Indonesian National Police Headquarters (e-MP), analyzing 8,831 cybercrime cases prosecuted in 2022.
- Sharma, 2024: Conducted content analysis on 40 reports generated via quasi-experiments to assess "Strength of Support" (SoS) categories.
- Hassan Shah Khan, 2025: Analyzed high-profile case studies including the 2014 Sony Pictures Hack, the 2021 Colonial Pipeline Ransomware Attack, and the 2018 Cambridge Analytical Data Scandal

In terms of the research articles focused on the advancements of digital forensics however prioritises simulation and pattern recognition over specific, localized datasets.

- **Data Sources:** Vaddi et al., 2024 emphasize the importance of diverse internet resources and training datasets for more accurate and precise findings. They specifically mention the use of unstructured crime report narratives and past records from external agencies Vaddi et al., 2024.

- **Visual Data:** Deep learning research focuses on forensic video analysis, specifically addressing low-quality CCTV footage. Techniques are devised to maximize the extraction of evidentiary elements through video enhancement Vaddi et al., 2024.
- **Heterogeneity:** Alenezi, 2023 notes that the dataset in smart environments is inherently heterogeneous, consisting of data from sensors, social media, and location-based services, which complicates correlation and reconstruction (Alenezi, 2023).

3.1.8 Results

The results of the research articles does reveals a discrepancy between theoretical standards and practical application:

- **Limitations of Reports:** Sharma, 2024 found that 82.5% of digital forensic studies were "Vague" in their methodology, and 0% explicitly asserted "Reliability" and "Validity" in a way consistent with other forensic sciences. This suggests a significant gap in academic and professional reporting standards.
- **Legal Admissibility:** Ilyas et al., 2024 concluded that while digital forensics is critical for finding "material truth," its success in Indonesia is hampered by a limited number of experts and internal police regulations that do not fully support the complexity of contemporary cybercrime.
- **Automation and AI:** Arif et al. (2025) and Hassan Shah Khan, 2025 both noted a rising trend in the use of AI and Machine Learning to automate evidence classification and anomaly detection, which helps in managing the "explosion of technology" and data volumes.

On the other hand, Table 5 showcases the resultant findings in regards to the possible advancements in terms of utilisation of Artificial Intelligence and Machine Learning:

Table 4: Comparative Results and Findings for AI and ML

Category	Vaddi et al., 2024	Alenezi, 2023
Detection Accuracy	High efficacy in predicting illegal behavior and risk detection	AI/ML improves the speed and accuracy of analyzing large, complex datasets
Data Integrity	Effectively prevents evidence loss or corruption in insecure systems (Blockchain)	Forensic readiness policies reduce the time and cost associated with incident response
Data Integrity	Effectively prevents evidence loss or corruption in insecure systems (Blockchain)	Forensic readiness policies reduce the time and cost associated with incident response
Video/Media	Deep learning allows for extraction of evidence from low-quality footage	AI, through training is able to recognize specific suspects via voice or facial patterns

3.1.9 Advantages and Limitations of Current Practices

The research is able to identify the benefits of standardised practices alongside the advancements of digital forensics while acknowledging significant barriers to universal implementation as well as the flaws of said advancement of digital forensics.

In regards to the Standardised Practices of Digital Forensics, the benefits and the flaws of its implementation are as follows:

■ Advantages

- **Consistency:** Frameworks like the NIST guidelines provide a "consistent method of data retrieval" Hassan Shah Khan, 2025).
- **Legal Integrity:** Adherence to chain-of-custody protocols ensures that evidence remains "legally solid" and admissible in court Ilyas et al., 2024.
- **Efficiency:** Standardised tools like Magnet AXIOM allow for cross-platform recovery (Cloud, Mobile, Computer), reducing the need for disparate systems.

■ Limitations

- **Encryption and Obfuscation:** Advanced encryption methods act as a primary barrier, preventing access to data despite standard protocols Hassan Shah Khan, 2025).
- **Jurisdictional Complexity:** Cybercrime is inherently global, and the lack of a harmonized legal frameworks makes cross-border evidence sharing difficult (Arif et al., 2025; Sharma, 2024).
- **Rapid Obsolescence:** The speed of technological change often outpaces the development of forensic standards, particularly in the IoT and cloud sectors Hassan Shah Khan, 2025).
- **Vague Reports:** As identified by Sharma, 2024, a lack of specific methodological reporting hinders the ability of practitioners to replicate findings or rely on published research in a legal context.

In regards to the Advancements of Digital Forensics, the benefits and the flaws of its implementation are as follows:

■ Advantages

- **Automation:** Both studies highlight that AI and ML automate time-consuming manual processes, such as the visual scrutiny of video footage or the parsing of massive log files Alenezi, 2023.
- **Transparency and Security:** The use of blockchain provides decentralization, removing the vulnerability of a "central authority" that might be targeted by attackers Vaddi et al., 2024.
- **Proactive Readiness:** Forensic readiness planning allows organizations to collect evidence in a "forensically sound manner" before an incident occurs (Alenezi, 2023).

■ Limitations

- **Technical Hurdles:** Vaddi et al., 2024 acknowledge the impossibility of an error-free framework, noting the persistence of false positives. Alenezi, 2023 highlights the challenges posed by encryption and the lack of standardization in communication protocols across different devices.

- **Ethical and Legal Constraints:** Both sources point to significant challenges regarding data privacy. Alenezi, 2023 specifically notes the difficulty in balancing the collection of personal data against the need to prosecute cybercrimes.
- **Resource Intensity:** Implementing these advanced frameworks (e.g., establishing a digital forensic laboratory or maintaining a blockchain network) requires substantial material and human resources (Alenezi, 2023; Vaddi et al., 2024).

3.2 Theme 2: AI Augmented Digital Forensics

The rapid evolution of cybercrime has necessitated a shift from traditional manual forensic techniques to AI-augmented digital forensics. As digital evidence proliferates across cloud platforms, encrypted services, and IoT devices, the volume and velocity of data have exceeded human processing capacities.

Through our gathered research articles, they highlight the integration of Artificial Intelligence (AI) not as a replacement for human judgment, but as a critical instrument for enhancing the speed, accuracy, and depth of cybersecurity investigations. (Sanna et al., 2025; Zubair et al., 2025).

3.2.1 Comparative Analysis of Research Focus and Methodologies

The provided research context explores AI's role in digital forensics (DF) through different lenses, ranging from theoretical procedural frameworks to specific algorithmic testing and defense against sophisticated threats.

3.2.2 Similarities in Research Perspectives

Across the studies, several consensus begin to emerge:

- **The Scalability Crisis:** Researchers agree that traditional forensic methods is incapable of keeping up with the terabytes of logs and chaotic evidence produced during enterprise breaches Zubair et al., 2025.
- **Human-Centric AI:** AI is framed as an "investigative partner" or "digital assistant" rather than a "black box" solution. Human supervision is deemed essential to validate findings and address the legal requirement for interpretability in court (Sanna et al., 2025; Zubair et al., 2025).
- **The Adversarial Nature of AI:** There is a shared recognition in that cybercriminals are leveraging AI for offensive purposes as well, such as automated intrusion, phishing, and anti-forensics techniques like steganography (SN et al., 2025; Sanna et al., 2025).

3.2.3 Methodological Differences and Research Goals

The researchers employ distinct strategies to address the advancement of the field through utilisation of AI, as demonstrated in Table 6.

Table 5: Comparative Results and Findings for AI and ML

Feature	Sanna et al., 2025	Zubair et al., 2025	SN et al., 2025
Primary Focus	Procedural integration across the DF pipeline (4 Phase Procedure)	Technical performance of hybrid deep learning model	Defense against Persistent Threats
Research Methodology	Qualitative framework proposal and LLM case studies	Quantitative testing using CNN-BiLSTM on synthetic datasets	Comprehensive literature and APT case studies
Key Entities	General cybercrimes, LLMs (Gemini, Copilot, ChatGPT)	Ransomware, DDoS, and Insider Threats	State-sponsored actors, Zero-day exploits
Core Argument	AI should be applied at all phases of digital forensics (From collection to reporting)	Hybrid models (CNN+Bi-LSTM) with attention are superior to standard ML	APTs require a multi-layered defense (Zero Trust)

3.2.4 AI Applications in the Digital Forensic Pipeline

The integration of AI is analyzed relative to the standard four-phase forensic procedure defined by NIST: Collection, Examination, Analysis, and Reporting.

1. Collection and Preservation

AI is able to facilitate the identification of digital evidence at crime scenes, through use of Computer Vision (CV) algorithms that can be deployed locally to assist first responders in identifying all relevant hardware (e.g., devices with volatile memory) to ensure all possible evidence has been recovered. AI also assists in terms of documenting the chain of custody by automatically recording and logging the device's conditions (Sanna et al., 2025).

2. Examination and Data Acquisition

In the examination phase, AI serves as a digital assistant by guiding consultants through technical procedures, based on reviewing similar past cases. This is particularly useful for recovering data from corrupted memory sections or damaged disks. (Sanna et al., 2025).

3. Analysis and Evidence Interpretation

This phase sees the most significant AI augmentation. Advanced models are used to:

- **Classify Multimedia:** Automated identification of specific data types (e.g., images, audio, chats) enhances privacy by only analyzing relevant data while searching for evidence. Besides that, it helps reduce potential trauma gain from analysts viewing sensitive and explicit materials (Sanna et al., 2025).
- **Pattern Recognition:** Deep learning models, specifically hybrid architectures like CNN-BiLSTM, excel at capturing both spatial dependencies in digital artifacts and temporal sequences in network traffic (Zubair et al., 2025).
- **Anti-Forensics Detection:** AI can be trained to detect hidden patterns in files used for steganography or to identify AI-generated "deepfake" content that mimics real individuals (Sanna et al., 2025).

4. Reporting and Legal Documentation Large Language Models (LLMs) and Natural Language Processing (NLP) are utilized to draft summaries and background technical definitions for legal parties. Future applications would include using NLP to review and score reports based on their understandability for non-technical stakeholders (Sanna et al., 2025).

3.2.5 Technical Performance and Model Robustness

Empirical testing suggests that advanced AI architectures significantly outperform traditional machine learning models in forensic tasks, as proven from Table 7 which uses Zubair et al., 2025 research of utilising a hybrid CNN-BiLSTM model with an attention mechanism and comparing it against baseline models such as Logistic Regression and Random Forest.

Table 6: Comparison of Model Metrics

Model Type	Accuracy/F1-Score	Time-to-Discovery	False Positiv
Logistic Regression	0.78	High (12+ seconds)	12.5%
Random Forest	0.82	Moderate (9 seconds)	9.8%
Standard LSTM	0.86	Moderate (7 seconds)	7.4%
CNN-BiLSTM + Attention	0.93 - 0.96 (AUC)	Low (4 seconds)	3.1%

3.2.6 Strengths of the Hybrid Approach

The CNN-BiLSTM model’s success is attributed to its **Attention Mechanism**, which assigns higher weight to important evidence segments. Ablation testing confirmed that removing attention decreased precision of the model, while removing metadata embeddings significantly decreased the model’s recall Zubair et al., 2025.

3.2.7 Challenges and Contradictory Findings

While AI offers substantial benefits, the article accurately identifies a few critical limitations and areas of friction within the implementation:

- The LLM Steganography Paradox Sanna et al., 2025 conducted a case study on popular LLMs (Gemini, Copilot, ChatGPT) regarding steganography.
 - Failure in Creation: None of the LLMs could successfully generate a working PNG image with steganographed content.
 - Success in Scripting: Conversely, the LLMs were highly proficient at generating functional Python code for encoding and decoding messages using Least Significant Bit (LSB) techniques.
 - Insight: This suggests LLMs are currently more useful as an offensive or defensive coding assistants rather than as direct content generators for anti-forensics.
- Attribution Complexity in APTs APTs (Advanced Persistent Threats) present a unique challenge due to their stealthy and state-sponsored nature. Attribution remains difficult because malicious actors use custom malware and fileless techniques

that runs only within the system memory instead of any other vulnerable areas. While AI improves threat hunting, the **automated attribution** of these actors remains a primary research gap (SN et al., 2025).

- **Data Bias and Environmental Variability** A significant weakness in AI forensic tools is the risk of biased models. Models that are trained on Western corporate data may risk failure when applied to mobile-centric systems in developing countries. Ensuring fairness and accuracy requires diverse, globally-sourced training datasets/ Zubair et al., 2025.

3.2.8 Emerging Trends and Future Directions

While reading through the articles, a trend can be observed in the fact that digital forensics is moving towards more resilient, transparent and collaborative ecosystems.

- **Explainable AI (XAI):** There appears to be a necessity for XAI to be properly utilised to provide clear justifications for the AI's own decisions. This transparency is vital for establishing trust among investigators and ensuring that the obtained evidence is admissible in court (SN et al., 2025; Zubair et al., 2025).
- **Blockchain Integration:** Researchers are exploring blockchain as a much more secure method for identity management, tamper-proof logging of forensic procedures alongside providing a decentralized threat intelligence sharing platform. (SN et al., 2025).
- **AI-vs-AI Feedback Loops:** A proposed trend involves a continuous loop where offensive AI develops evasive techniques to better test defensive AI's responses. Human supervisors are required to monitor this interaction to better refine both of the AI's detection algorithms and analyst skills (Sanna et al., 2025).
- **Federated Learning:** This technique allows organizations to collaboratively train APT detection models without sharing sensitive private data, thereby preserving privacy while increasing the shared knowledge base (SN et al., 2025).

3.3 Theme 3: Cybercrime and the Detection of Digital Deceptions

3.3.1 Overview of the Cybercrime Landscape

The global cost of cybercrime is projected to reach \$10.5 trillion by 2025 Osamor et al., 2026. This financial escalation is driven by the increasing sophistication of digital deception, primarily through two vectors: the democratization of high-fidelity synthetic media (deepfakes) and the refinement of traditional phishing via spoofed communications. Both methods exploit human trust and systemic vulnerabilities in standard authentication protocols.

3.3.2 Comparative Analysis of Deception Techniques

1. Deepfake Generation and Proliferation

According to Singh and Dhumane, 2025, deepfakes utilize Generative Adversarial Networks (GANs) to create hyper-realistic images, audio, and video through

techniques like face synthesis (e.g., StyleGAN), face morphing, and speech cloning (e.g., Wav2Lip). This could be explained through the fact that open-source tools like DeepFaceLab and Faceswap have lowered the barrier of entry, allowing non-technical actors to generate deceptive content Singh and Dhumane, 2025. No doubt the high prevalence of deepfakes like the Deepfake cases in India alone rose by 550%, with 90% of surveyed individuals in the UK expressing apprehension about the technology Singh and Dhumane, 2025.

2. Email Spoofing and Phishing

Osamor et al., 2026 highlight that phishing remains a dominant threat, with a 1726% increase in phishing websites between 2011 and 2021. Cybercriminals are able to use tools like malicious IP addresses, forged email headers, and tracking pixels which are used to monitor the recipient’s activity and location as an invasion of their privacy.

Unlike deepfakes, which often rely on visual/auditory realism, email spoofing instead leverages social engineering triggers such as emergencies and sudden authority presence (e.g., "Microsoft account unusual sign-in activity") to bypass the victim’s critical thinking Osamor et al., 2026.

3.3.3 Detection Methodologies

To better combat the cybercrimes, researchers have propose diverse approaches to identifying digital deceptions, ranging from automated AI models to manual forensic analysis, as can be seen in Table 8.

Table 7: Detection Techniques and Capabilities

Technique	Application	Capabilities	Limitations
CNN-Based Models	Deepfakes (Image/Video)	High accuracy in spatial artifacts Singh and Dhumane, 2025	Sensitive to compression resolution
RNN/LSTM Models	Deepfakes (Image/Video)	Captures temporal inconsistencies and motion patterns Singh and Dhumane, 2025	High computation inference
Header Analysis	Email Spoofing	Identifies mismatches in SPF, DKIM, and DMARC Osamor et al., 2026	Can be bypassed sophisticated obfuscation
Biometric Defenses	Identity Verification	Liveness detection (eye blinks, pupil dilation) and 3D depth sensing Singh and Dhumane, 2025	Subject-dependent input quality
Hybrid Approaches	Multi-modal Detection	Combines human-in-the-loop verification with AI-based filtering Osamor et al., 2026	Complex design resource-heavy

1. Similarities in Research Findings Both studies emphasize that **authentication failures** are primary indicators of fraud. In email forensics, this manifests as **spf=none** or **dkim=none** results Osamor et al., 2026. In deepfake detection, this involves identifying "spoofing" where synthetic data is presented as genuine biometric data Singh and Dhumane, 2025. Furthermore, both identify **demographic**

and environmental biases in training data (e.g., FFHQ and DFDC datasets) as a significant hurdle for universal detection Singh and Dhumane, 2025.

2. Differences and Contradictory Findings

- **Scalability:** Singh and Dhumane, 2025 note that heavy AI pipelines (Transformers) are computationally expensive and struggle with real-time requirements (running at 18–22 fps). Conversely, Osamor et al., 2026 argue for the continued necessity of manual forensic analysis, asserting that automated tools often overlook subtle anomalies like a 2-second timestamp mismatch in email hops.
- **Human Factor:** Osamor et al., 2026 emphasises human behavior as a **critical vulnerability**, whereas Singh and Dhumane, 2025 focuses more on the technical **arms race** between generative models and detection algorithms while arguing for the human element for the detector models.

3.3.4 Strengths, Weaknesses, and Vulnerabilities

1. Technical Weaknesses

- **Cross-Dataset Generalization:** Models trained on specific datasets (e.g., DFDC) often fail when tested against "in-the-wild" content. For example, the accuracy of the models can drop from 90% to 60% when facing user-generated content like compressed TikTok videos Singh and Dhumane, 2025.
- **Adversarial Attacks:** Attackers can add **intentional perturbations** (noise) to forged media to deceive CNN-based classifiers into labeling them as authentic Singh and Dhumane, 2025.

2. Forensic Failures Real-world incidents demonstrate the stakes of these vulnerabilities:

- **Political Turmoil:** The 2019 Ali Bongo incident in Gabon showed how a suspected deepfake (even if genuine) can trigger military takeover attempts due to public mistrust Singh and Dhumane, 2025.
- **Financial Loss:** A 2019 voice deepfake attack against a UK-based CEO resulted in a \$243,000 fraudulent wire transfer, illustrating the danger of voice cloning in high-stakes environments Singh and Dhumane, 2025.

3.3.5 Emerging Trends and Policy Frameworks

- **Research Trends** While analysing the research articles for this review, certain trends tend to pop up. One in particular being the Explainable AI (XAI), which are models that provide reasonings alongside an interpretable output to support legal admissibility and public trust in their usage Singh and Dhumane, 2025. Another one would be the facilitation of Federated Learning, a privacy-preserving approach where models are able to be trained across devices without centralizing raw biometric data Singh and Dhumane, 2025. In addition, there seems to be talks of the implementation of continuous authentication: a system that monitor user behaviors (gait, keystroke rhythm) throughout a whole session rather than relying on a

one-time login Singh and Dhumane, 2025, which could better ensure the user isn't a malicious actor that pretends to be another user.

- **Global Regulatory Initiatives** In terms of global efforts in the other hand, it appears that there are new legal frameworks being established, with the aim to categorize and mitigate AI risks. For one, the EU AI Act (2024) has categorised deepfakes as **high-risk**, requiring transparency measures such as labeling synthetic content Singh and Dhumane, 2025 to avoid misconception. As observable from NIST's Incident Response, Osamor et al., 2026 advocates for integrating the NIST Incident Response Lifecycle with current forensic investigations to standardize how organizations should handle spoofing attacks. In addition, a brand new strategy has appeared: **Provenance Tagging**, which uses cryptographic hashes to track the origin of digital media, is currently being supported by the Partnership on AI (2023) to identify tampered content Singh and Dhumane, 2025.

4 Comparative Analysis of Existing Studies

Provide a systematic comparison of the reviewed research articles.

A table such as the following is recommended.

Table 8: Comparative analysis of the reviewed research studies

No.	Author & Year	Research Objective	Method/Technique	Dataset/Sample	Key Findings	Limitations
1	[Author, Year]	[Research objective]	[Method/Technique]	[Dataset/Sample]	[Key findings]	[Limitations]
2	[Author, Year]	[Research objective]	[Method/Technique]	[Dataset/Sample]	[Key findings]	[Limitations]
3	[Author, Year]	[Research objective]	[Method/Technique]	[Dataset/Sample]	[Key findings]	[Limitations]
4	[Author, Year]	[Research objective]	[Method/Technique]	[Dataset/Sample]	[Key findings]	[Limitations]

5 Synthesis of Findings

This section evaluates the collective evidence and thematic patterns retrieved from the reviewed literature.

5.1 What Do the Studies Agree On?

The analyzed literature demonstrates broad consensus across multiple key areas regarding modern cybercrime and digital forensics:

1. **Digital forensics is essential for fighting cybercrime:** Available research consistently indicates that digital forensics is vital for investigating and prosecuting cybercrimes. As information systems become more prevalent, criminal activity increasingly occurs within digital environments. Digital forensics assists law enforcement officers in locating, examining, and protecting electronic data to identify suspects and establish legal facts in court.
2. **Traditional methods are insufficient:** There is a strong consensus that traditional manual forensic techniques are no longer adequate for contemporary cybercrime. Data volumes are too large for manual processing alone, and perpetrators employ sophisticated methods such as encryption and anti-forensic software to conceal activities. Manual processes are also too slow, meaning security breaches often cause harm before investigators can identify the event.
3. **AI and machine learning are necessary upgrades:** Various studies suggest that artificial intelligence, machine learning, and deep learning provide necessary methods to address current forensic difficulties. AI systems analyze large datasets rapidly, identify configurations that human investigators might overlook, and automate repetitive duties. Rather than replacing human specialists, AI enables professionals to operate with greater speed and precision.
4. **Encryption and anti-forensics pose major obstacles:** The literature identifies encryption and anti-forensic methods as significant barriers. Encryption secures data to prevent investigators from viewing content, while tracking pixels and forged emails help malicious actors evade standard security filters. Additionally, AI software can sometimes generate steganographic code to mask hidden information.
5. **Zero Trust and layered defenses work best:** To address these risks, multiple studies propose utilizing Zero Trust Architecture—operating on a “never trust, always verify” principle—as an effective protection mechanism against complex cyber threats and deceptive email campaigns.
6. **Standardization is urgently needed:** The research highlights a clear lack of standardized protocols across the field of digital forensics. Different investigators frequently apply divergent methodologies, which often leads to inconsistent analytical outcomes.

5.2 What Do the Studies Disagree On?

The literature reveals notable divergences and contrasting conclusions among researchers regarding specific technologies and methodologies:

1. **Should AI be explainable?:** Researchers in multiple publications state that artificial intelligence must be explainable so that judges and law enforcement officers can trust and interpret software results. Conversely, other scholars argue that system precision and speed are significantly more important than logic clarity, leaving no consensus among academics.
2. **Is blockchain a good forensic solution?:** Certain studies describe blockchain as a permanent, tamper-resistant ledger for evidence. In contrast, other reports characterize blockchain as slow, capital-intensive, and electricity-consuming, creating a persistent disagreement.
3. **Is generative AI a threat or a tool?:** Some reports suggest generative AI is hazardous because it produces fabricated audiovisual media and assists cybercriminals. Meanwhile, other researchers demonstrate that the exact same technology serves to identify and block those fabricated files, indicating a split perspective.

5.3 Which Approaches Appear to Be Most Effective?

Comparative evaluation of the reviewed literature highlights several standout methodologies:

1. **Hybrid deep learning models:** The CNN-BiLSTM with Attention model outperforms alternative computational processes, achieving a correct identification rate of 94.8% while generating minimal false alarms and extracting digital proof in roughly 4 seconds (compared to 9 to 12 seconds for other systems).
2. **Entropy-weighted federated learning:** This methodology utilizes uncertainty calculations to select reliable data origins, attaining a 94.8% success rate while maintaining a minimal frequency of erroneous identifications.
3. **Multimodal detection systems:** Integrating visual, audio, and text-based signals represents the most efficient strategy for deepfake identification, as it recognizes diverse data alteration categories and functions robustly across varied collections of information.
4. **NIST framework integration:** Applying the NIST Incident Response framework alongside physical forensic examination successfully analyzes phishing attacks, revealing authentication failures and hidden tracking anomalies.

5.4 What Trends Can Be Observed?

Emerging patterns across the literature span multiple technical domains:

1. **AI is used at every forensic stage:** Artificial intelligence applications extend from computer vision identifying hardware during evidence gathering, to large language models acting as digital assistants, CNN-BiLSTM models processing data, and natural language processing generating formal reports.
2. **Federated learning is growing:** Federated learning allows organizations to train models while source data remains in place, satisfying privacy regulations and protecting confidential records during cross-border investigations.
3. **Automated attribution is emerging:** Security analysts leverage AI to link digital intrusions to specific attackers by identifying recurring hostile behaviors, shortening the window between breach detection and perpetrator identification.
4. **Deepfake detection is a priority:** Escalating incidents involving deepfake media are driving researchers to develop more sophisticated detection models and biometric protection systems that function reliably across diverse datasets.
5. **Cloud and IoT forensics are expanding:** Investigations involving cloud services and IoT devices require specialized tools, cross-border legal frameworks, and organizational forensic readiness to handle complex operational incidents.

6 Research Gaps

Building upon the findings and themes established in Section ??, several critical research gaps remain unaddressed in the existing literature:

1. **No standard forensic protocols:** The majority of digital forensic studies—specifically 82.5%—are “vague” in their methodology. In the absence of standardized procedures, forensic investigations lack consistency, and legal authorities may exclude evidence from judicial proceedings.
2. **Detection models don’t work well in real-world conditions:** The performance of models trained on a specific dataset drops significantly when processing different data types. It is common for a model to achieve over 90% accuracy in a laboratory setting, while accuracy decreases to approximately 60% on data from actual environments, compounded by demographic biases that cause non-uniform performance.
3. **AI tools can’t explain their decisions:** AI models frequently function as “black boxes” that produce decisions without providing explanations for their underlying logic. Because courts of law require transparent and verifiable evidence, judges may refuse to accept AI-generated outputs.
4. **No cross-domain frameworks:** Many systems are restricted to a single environment (such as the Internet of Things, cloud services, or corporate networks) and fail to interoperate. Because cybercrime frequently spans multiple domains, investigators find it difficult to assemble evidence seamlessly from diverse sources.
5. **Weak defense against anti-forensics:** Malicious actors use Large Language Models to generate steganographic code, employ tracking pixels to bypass filters, and utilize encryption to prevent access. Criminals deploy advanced obfuscation methods that current forensic software cannot reliably identify.
6. **Limited real-time detection:** Transformer models currently operate at only 18 to 22 frames per second, which is insufficient for real-time applications. Furthermore, edge devices frequently lack the processing power required to support modern AI models.
7. **Jurisdictional and cross-border issues:** Due to conflicting international laws, varied privacy regulations, and jurisdictional disputes, investigations crossing borders remain highly complex. Cybercrime operates on a global scale, whereas legal frameworks remain strictly national.
8. **Weak multimedia forensics for emerging threats:** The detection of diffusion-based fakes, real-time deepfakes, and AI-generated synthetic media is still in early development. New AI tools produce realistic fake content that current systems are unable to identify reliably.

7 Future Research Directions

To fix all those issues we found in Section ??, here are some directions future research should probably take:

1. **Make AI explainable:** Since judges won't take black-box evidence, people really need to build AI that actually shows its work and explains its logic clearly to investigators.
2. **Create better datasets:** Researchers have to stop relying on tiny lab sets and build bigger datasets with actual real-world conditions, varied environments, and different demographics so accuracy doesn't plummet to 60%.
3. **Build cross-domain frameworks:** Someone needs to design tools with standard APIs that can pull and share evidence across IoT, cloud, and corporate networks instead of keeping everything trapped in silos.
4. **Create AI-vs-AI feedback systems:** Since criminals are using LLMs for anti-forensics, we need defensive AI that constantly tests itself against automated attacks and updates on the fly.
5. **Make lightweight, real-time solutions:** Transformer models are way too slow right now, so future work has to compress these models or split processing so they can run smoothly on edge devices for live detection.
6. **Use federated learning for privacy:** To get around cross-border privacy laws, studies should look into federated learning and maybe blockchain so teams can share threat intel without exposing sensitive raw data.
7. **Harmonize legal frameworks:** We need people to work on global cybercrime treaties and standard certifications so investigators aren't constantly blocked by conflicting national laws and vague protocols.
8. **Improve multimedia forensics:** Technology is moving fast with diffusion fakes and synthetic media, so forensic tools need major upgrades, like live video analysis and cryptographic checks, to actually catch these new tricks.

8 Discussion

Provide an overall critical discussion of the literature.

The discussion should answer:

- What has the research community achieved so far?
- Which approaches are most promising?
- What are the major challenges?
- Are the current approaches sufficient?
- What limitations remain?

- How has the research area evolved?
- What important issues remain unresolved?

This section should demonstrate **critical thinking**, not merely summarisation.

9 Conclusion

This review aimed to explore the role of digital forensics in cybercrime investigations, focusing on how artificial intelligence is changing the field and how well current methods handle the challenges of modern cybercrime. From over thirty articles, three main themes emerged: the current state of standardized forensic practices, the increasing use of AI in forensics, and the ongoing battle between digital deception and detection. Standard forensic methods still support evidence collection but face issues like encryption, different laws across regions, fast-changing technology, and inconsistent reporting.

AI-enhanced methods, especially advanced models like CNN-BiLSTM, have shown better performance than traditional methods in accuracy and speed, while also automating tasks like sorting media and finding anomalies. AI is seen as a tool to help human investigators, not replace them, because human checks are still needed to meet legal standards. The issue of cybercrime and digital trickery shows how fake media and communications exploit human trust and tech weaknesses, with different detection methods having their own pros and cons.

The studies highlight major gaps: a lack of diverse global datasets, AI models that are hard to explain, no unified international laws for sharing digital evidence, and difficulties in tracing complex attacks by state actors. Despite progress, digital forensics still needs a clear, standardized, and legally strong approach to tackle cybercrime effectively. Moving forward, combining traditional methods with AI, focusing on explainable AI, using federated learning, ensuring evidence integrity with blockchain, and developing secure cryptographic techniques are key to keeping up with the evolving cybercrime landscape.

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APPENDIX A – INDIVIDUAL ARTICLE REVIEW

Student: Name / ID

Table 9: Individual article review

Article	Research Problem	Objective	Methodology	Dataset/Sample	Key Findings	Limitations	Research Gap
xx & xx	xx	xx	xx	xx	xx	xx	xx
xx & xx	xx	xx	xx	xx	xx	xx	xx
xx & xx	xx	xx	xx	xx	xx	xx	xx

APPENDIX A – INDIVIDUAL ARTICLE REVIEW

Student: Name / ID

Table 10: Individual article review

Article	Research Problem	Objective	Methodology	Dataset/Sample	Key Findings	Limitations	Research Gap
xx & xx	xx	xx	xx	xx	xx	xx	xx
xx & xx	xx	xx	xx	xx	xx	xx	xx
xx & xx	xx	xx	xx	xx	xx	xx	xx

APPENDIX A – INDIVIDUAL ARTICLE REVIEW

Student: Name / ID

Table 11: Individual article review

Article	Research Problem	Objective	Methodology	Dataset/Sample	Key Findings	Limitations	Research Gap
xx & xx	xx	xx	xx	xx	xx	xx	xx
xx & xx	xx	xx	xx	xx	xx	xx	xx
xx & xx	xx	xx	xx	xx	xx	xx	xx

APPENDIX A – INDIVIDUAL ARTICLE REVIEW

Student: Nur Afiqah Ramli / 251UC2514E

Table 12: Individual article review

Article	Research Problem	Objective	Methodology	Dataset/Sample	Key Findings	Limitations	Research Gap
Sanna, Regano, Maiorca & Giacinto (2025)	Traditional digital forensics struggles with complex anti-forensics like steganography and high data volumes.	To map AI techniques to the NIST 4-phase DF process and test LLM capabilities in handling steganography.	Qualitative framework proposal combined with empirical case studies using Gemini, Copilot, and ChatGPT-4o.	Steganographic images and functional Python scripts generated by LLMs; GitHub datasets.	AI enhances all 4 DF phases. LLMs all utilized LSB but failed to decode images via upload; however, they successfully generated working encoding/decoding scripts.	Small scope; no automated countermeasures proposed for the generated threats.	Lack of automated countermeasures and comprehensive testing across a wider variety of anti-forensic techniques.
Alenezi (2023)	Retrieving and preserving digital evidence across decentralized smart environments is highly complex due to lack of standardization.	To survey recent advancements and challenges in retrieving and preserving evidence across IoT and connected devices.	Comprehensive literature survey and categorization of smart environment forensics.	Recent literature and state-of-the-art tools focusing on smart environments and IoT.	Identified 4 advancement categories and 6 challenge categories (including data heterogeneity and legal issues). AI/ML improves processing speed but faces standardization hurdles.	Broad scope lacks deep empirical data; limited depth in specific IoT and cloud scenarios.	Need for standardized protocols for smart environments and empirical validation of proposed forensic readiness frameworks.
Vaddi, Kamble, Vaingankar, Khatri & Bhalerao (2024)	The rapid evolution of cybercrime necessitates intelligent tools to handle modern threats across diverse, complex domains.	To review digital forensic evolution over 5 years focusing on AI, blockchain, cloud, and machine learning to guide developers.	Literature review and statistical analysis of long-standing and recent techniques.	Unstructured crime report narratives and literature patterns post-2018.	AI, blockchain, and deep learning drastically improve forensic capabilities, particularly in cloud/IoT architectures, ensuring secure chain of custody.	Specific empirical testing details and localized datasets are limited in the review scope.	Bridging the gap between theoretical frameworks and real-world deployment while reducing false positive rates in automated systems.

APPENDIX B – ARTICLE MATRIX

Table 13: Group article matrix of reviewed research studies

No.	Article	Student	Method	Dataset	Key Results	Strength	Limitation	Research Gap
1	[Insert Paper]	Wan Wei Siang						
2	[Insert Paper]	Wan Wei Siang						
3	[Insert Paper]	Wan Wei Siang						
4	[Insert Paper]	Elsa Zara Binti Fakhurrazi						
5	[Insert Paper]	Elsa Zara Binti Fakhurrazi						
6	[Insert Paper]	Elsa Zara Binti Fakhurrazi						
7	[Insert Paper]	Terry Teoh Jay Poh						
8	[Insert Paper]	Terry Teoh Jay Poh						
9	[Insert Paper]	Terry Teoh Jay Poh						
10	Sanna, Regano, Maiorca & Giacinto	Nur Afiqah Ramli	Qualitative framework proposal combined with empirical case studies using Gemini, Copilot, and ChatGPT-4o.	Steganographic images and functional Python scripts generated by LLMs; GitHub datasets.	AI enhances all 4 DF phases. LLMs all utilized LSB but failed to decode images via upload; however, they successfully generated working encoding/decoding scripts.	Practical evaluation of multiple prominent LLMs handling steganography.	Small scope; no automated countermeasures proposed for the generated threats.	Lack of automated countermeasures and comprehensive testing across a wider variety of anti-forensic techniques.
11	Alenezi	Nur Afiqah Ramli	Comprehensive literature survey and categorization of smart environment forensics.	Recent literature and state-of-the-art tools focusing on smart environments and IoT.	Identified 4 advancement categories and 6 challenge categories (including data heterogeneity and legal issues). AI/ML improves processing speed but faces standardization hurdles.	Broad scoping across multiple decentralized smart environments and IoT layers.	Broad scope lacks deep empirical data; limited depth in specific IoT and cloud scenarios.	Need for standardized protocols for smart environments and empirical validation of proposed forensic readiness frameworks.
12	Vaddi, Kamble, Vaingankar, Khatri & Bhalerao	Nur Afiqah Ramli	Literature review and statistical analysis of long-standing and recent techniques.	Unstructured crime report narratives and literature patterns post-2018.	AI, blockchain, and deep learning drastically improve forensic capabilities, particularly in cloud/IoT architectures, ensuring secure chain of custody.	5-year evolution window focusing heavily on emerging tech like blockchain and cloud.	Specific empirical testing details and localized datasets are limited in the review scope.	Bridging the gap between theoretical frameworks and real-world deployment while reducing false positive rates in automated systems.